Tzu-Ching Liao

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Summary

Data Science MS candidate at USC designing LLM–orchestrated multi-model systems that route time series tasks across domains(healthcare, finance, transportation) to specialized forecasting modes.

Education

University of Southern California, Master of Science in Applied Data Science — Aug. 2025 – May. 2027

National Taipei University, Bachelor of Business Administration in Statistics — Sep. 2020 – Jun. 2024

Professional Experience

Research Intern — Mar. 2026 - Present

Machine Learning and Data Mining Lab, Department of Computer Science, USC

- Contributed a new foundation model **(Datadog Toto2)** to an time series ensemble framework designed for future **LLM agent**, implementing a GluonTS-based adapter (lazy model loading, OOM-safe batching) conforming to the framework's **unified forecast/cross-validation interface**.
- Built a cross-family benchmark unifying evaluation of 28 models — spatio-temporal GNNs, zero-shot foundation models, trained SSMs, and classical baselines — for reproducible large-scale traffic forecasting across LargeST's road networks (716–8,600 sensors).
- Led the design of the training and evaluation infrastructure, including fault-tolerant HPC training orchestration and per-model dependency isolation, and fixed an evaluation inconsistency that had understated cross-family error by up to 31%, ensuring the benchmark's results held up under scrutiny.

Data Engineer Intern — Jan. 2025 - Jun. 2025

SYSTEX - specializing in cloud services, AI, and integrated IT ecosystems, Taipei, Taiwan

- Architected a **high-velocity SQL Lakehouse** on **AWS** using Glue and PySpark , integrating automated schema validation and **CI/CD** pipelines for S3 infrastructure management to ensure data integrity. This optimization slashed data-to-analysis **latency** by **70%** , enabling the business to shift from reactive reporting to real-time, proactive decision-making.
- Boosted executive decision efficiency by **50%** by architecting low-latency data pipelines for live Tableau dashboards, replacing static reporting with real-time interactive analytics to enable faster strategic pivots.

Research Assistant — Jul. 2022 – Jun. 2023

Department of Statistics, National Taipei University — **Advisor: Prof. Hong-Long Wang**

- Accelerated retail strategy optimization in a growing market by engineering a high-scale predictive framework **(XGBoost/MLP)** on **3.5 billion records** , achieving **86%** Recall for **repurchase forecasting**.

Project Experience

Conversational Time Series Models — LLM/Agentic-Based Time Series Forecasting — Jul. 2026 – Present

Fork of TSOrchestra (DC-research/TSorchestra); pull request in progress

- Built the tool-use layer for an **agentic forecasting system**: designed a **uniform adapter interface** that exposes 5+ time-series foundation models (Toto 2.0, TiRex, TimesFM 2.5, ...) as interchangeable actions behind one contract, so the selection policy can be swapped without touching any model integration.
- Designed the **constrained-optimization layer** of an agentic ensemble controller: **SLSQP** fits non-negative, sum-to-one weights across 5 interchangeable forecast-accuracy objectives on rolling-origin cross-validation (avoiding test leakage), with equal-weight fallback on solver non-convergence.

Traffic Forecasting Model Evaluation Benchmark

- Designed a **unified, config-driven benchmark framework** integrating the **evaluation harness from GIFT-Eval** (a benchmark for general time series forecasting model evaluation), letting **25+ forecasting models** (**GNNs**, **foundation models**, **SSMs**, and classical baselines) share one evaluation interface and produce 250+ reproducible evaluation logs.
- Designed a **fault-tolerant checkpoint/resume system** that auto-detects and re-queues unfinished HPC training jobs across the cluster's 48-hour wall-clock limit, automating **103** job continuations (34 from timeouts) with **zero manual monitoring** — distinguishing recoverable failures (timeout, node failure) from fatal ones (OOM) to avoid futile retries.

Publications

- D. Cao, T. Liao, Y. Xu, N. Ghasem, Q. Guo, J. Liu, M. Weng, T. Hirano, and Y. Liu. "Traffic Forecasting in Practice: Comparing Graph Models and Foundation Models for Deployment-Aware Traffic Forecasting." *Under Review*, 2026.